

MUHAMMAD ROSHAAN

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OBJECTIVE

Final-year Computer Science student with hands-on experience building high-concurrency backend systems, RESTful APIs, and data-driven pipelines. Proficient in Python, FastAPI, SQL/NoSQL databases, and Power BI, with a proven ability to design systems that eliminate race conditions and deliver reliable, scalable performance under load. Seeking an internship or junior role in backend engineering or data analytics to apply strong system design and analytical skills in production environments.

EDUCATION

Sir Syed University of Engineering and Technology, Karachi

Sep 2023 – Jul 2027

BS in Computer Science | CGPA: 3.81/4.00 (6th Semester)

Relevant Coursework: Python, Software Engineering, Data Structures, Database Management Systems, OOP, Operating Systems

TECHNICAL SKILLS

Backend & Architecture: Python (Advanced), FastAPI, Celery, RESTful API Design, SQLAlchemy ORM, WebSockets, JWT Authentication

Concurrency & Infrastructure: Redis (Pub/Sub, Distributed Locks, Caching), RabbitMQ, Pessimistic Locking, Asynchronous Worker Queues, Docker

Databases & Cloud: MySQL, PostgreSQL, SQLite, MongoDB, Supabase (Real-Time Sync)

Data Engineering & BI: Pandas, NumPy, Matplotlib, ETL Pipelines, Power BI, DAX

Core Competencies: High-Concurrency System Design, Race Condition Elimination, Zero Data-Loss Pipelines, System Stability Under Load

PROFESSIONAL EXPERIENCE

Data Science Intern - 10Pearls Pakistan (NASTP)

Apr 2026 - Jun 2026

10P Shine Internship Program - Remote

- Built end-to-end ML pipelines, from real-time data ingestion and feature engineering in Hopsworks to trained model deployment.
- Trained and deployed an AQI (Air Quality Index) prediction model, applying MLOps practices in a production setting.

Data Processing Engineer - Softix Solution

Apr 2026

Full-Time - On-site

- Handled ticket purchasing through Ticketmaster and processed incoming orders, maintaining accurate and up-to-date database records.

PROJECTS

SeatVault - Transaction-Safe Concurrency Engine

Mar 2026 – Apr 2026

FastAPI + Redis + Celery + MySQL + React

- Engineered a high-concurrency ticketing system using Redis distributed locks and FastAPI to prevent double-booking and race conditions during flash sales.
- Integrated Celery for asynchronous background processing and built a React frontend with real-time polling and JWT authentication.

Enterprise ETL & Inventory Pipeline

July 2026

FastAPI + Pandas + SQLAlchemy + Supabase(PostgreSQL) + Celery + Redis

- Built an automated ETL pipeline using Pandas and SQLAlchemy to clean and bulk-load 540K+ records into Supabase PostgreSQL.
- Integrated Celery and Redis for automated background scheduling and real-time inventory threshold alerts.

FlashSale Concurrency Engine - Inventory Integrity System

Oct 2025

FastAPI + MySQL

- Designed a high-concurrency backend using pessimistic locking within SQLAlchemy database transactions to eliminate race conditions and prevent overselling in inventory management.

BooknScore - Offline-First Scoring Database

Dec 2025 – Mar 2026

Flutter + Supabase(PostgreSQL) + SQLite

- Designed the relational schema and SQLite scoring engine from the ground up, then layered real-time Supabase cloud synchronization on top to guarantee zero data loss during network interruptions.

CERTIFICATIONS & COURSES

Introduction to MongoDB in Python - DataCamp (Nov 2025): MongoDB fundamentals, NoSQL concepts, and core database operations in Python.

FastAPI (Python) - Udemy (Jul – Oct 2025): Built and deployed high-performance RESTful APIs with database integration, validation, and modular design (21.5 hours).

LANGUAGES

English: Fluent - proficient in technical writing and professional communication.

Urdu: Native - primary language for everyday communication.